Assignment: Different Kernels

Task-1:

1. Consider the following grayscale input image ***I*** of size 5×5:

$$I = \begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 4 & 1 & 2 & 0 & 1 \\ 1 & 3 & 1 & 2 & 2 \\ 0 & 2 & 4 & 1 & 0 \\ 1 & 1 & 2 & 3 & 1 \end{bmatrix}$$

   A convolution layer uses the following 3×3 kernel:

$$K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

   A. Calculate the output size after performing the convolution operation with Stride = 1 and Padding = 0.
   B. Compute every element of the convolution output feature map. Show all intermediate calculations.
   C. Compute the final pooled feature map and its size after applying a 2×2 max-pooling operation with stride 2.
   D. Calculate the total trainable parameters of the convolution layer, including biases when Input channels = 1, Number of filters = 10, and Kernel size = 3×3.
   E. If padding is changed from P = 0 to P = 1:
      I. What will be the new output size?
      II. Explain why the output size changes.
   F. How many multiplications are required to compute one output pixel for one filter?
   G. How many multiplications are required to compute one output pixel for one filter?
   H. How many multiplications are required if there are 10 filters?

2. Assume the input feature map has size 4 × 4 × 2 (Height × Width × Channels):

3.

$$X_1 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} \quad X_2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \end{bmatrix}$$

Also assume a Convolution Kernel (2 × 2 × 2):

$$K_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad K_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Perform the following operations and compute the resulting feature map size (and values where necessary):

A. Regular convolution without zero padding and with stride = 2
B. Regular convolution with zero padding and with stride = 2
C. Depthwise convolution without zero padding and with stride = 1

Clearly show all steps, including:

a. Output dimension calculations
b. Intermediate computations
c. Final feature maps

4. Assume the convolution output contains the value −5. Compute the output when using: ReLU, Sigmoid, and Tanh.

Task-2:

Design and implement a **Binary CNN-based classifier** to classify images into **your favourite two classes** (e.g., cats vs. dogs, apples vs. oranges, flowers vs. leaves).

Conditions

1. **Dataset:**
   - Capture all images using your **own mobile phone**.
   - Use approximately the same number of images for both classes.
   - Clearly separate the images into the two classes.

2. **CNN Models:**
   Design and train classifiers using the following types of convolution:
   - Regular convolution kernel
   - Pointwise convolution kernel (1 × 1)
   - Deformable convolution kernel
   - Atrous/Dilated convolution kernel
   - Depthwise separable convolution kernel

3. **Comparison:**
   Compare the performance of the five CNN-based classifiers using appropriate evaluation metrics, such as:
   - Accuracy
   - Precision
   - Recall
   - F1-score
   - Confusion matrix

4. **Analysis:**
   Discuss:
   - Which convolution type provides the best classification performance?
   - Which model requires the least computational resources?
   - How do the different convolution types affect feature extraction?
   - Why might one convolution type perform better than another for your chosen classes?

5. **Report:**
   Your report should include:
   - Selected two classes and motivation
   - Image acquisition procedure
   - Dataset size and train/validation/test split
   - CNN architecture
   - Training procedure
   - Results and comparison
   - Confusion matrices
   - Discussion and conclusion

Important: Keep the **CNN architecture, training settings, and dataset split as similar as reasonably possible** across the five experiments so that the comparison focuses on the effect of the convolution type.